

Attention Deficit Disorder and the Mediating Effect of Age on Academic and Behavioral Variables

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ABSTRACT. Seventy-nine children, independently diagnosed as having attention deficit disorder (ADD) with or without hyperactivity and ranging in age from 6 years, 0 months, to 12 years, 11 months, were administered the Wechsler Intelligence Scale for Children-Revised (WISC-R), the Wide Range Achievement Test (WRAT), and the Child Behavior Checklist (CBC). Multiple regression analyses indicated a significant multiple correlation between age and achievement variables ($R = 0.42$, $p = 0.003$) and age and behavioral (CBC) variables ($R = 0.55$, $p = 0.01$). Post hoc analyses indicated that much of this variance was accounted for by a significant negative association between age and Full Scale IQ-WRAT Arithmetic difference scores, and significant positive correlations between age and scores on the Social Withdrawal and Uncommunicative scales from the CBC. A nonparametric chi-square analysis indicated that older children with ADD are significantly more likely ($p = 0.037$) than younger ADD children to have a discrepancy of 15 or more points between IQ and math achievement scores. Results suggest that older ADD children are more likely than younger ADD children to experience academic and socioemotional difficulties. *J Dev Behav Pediatr* 11:22-26, 1990. Index terms: attention deficit, hyperactivity, age, learning disability.

Attention deficit disorder (ADD) is a developmental disorder that is characterized by impulsivity, distractibility, and inattention, and may be accompanied by symptoms of hyperactivity. The nomenclature for this disorder has recently been changed to attention deficit hyperactivity disorder in the revised version of the Diagnostic and Statistical Manual for Mental Disorders (DSM-III-R).¹ Although the characteristics of the disorder have remained essentially the same, the distinction between ADD with hyperactivity and ADD without hyperactivity has been eliminated.

Estimates for the prevalence of ADD vary; however, using a conservative approach, it is estimated that 2-3% of the childhood population have this disorder.^{1,2} This translates to approximately 1.2 million school-age children in the United States.

In terms of etiology, one hypothesis is that ADD symptoms may be related to a maturational lag in the development of attentional processes and inhibitory mechanisms.³⁻⁵ To investigate this hypothesis, we recently examined the relationship between hyperactivity and neuropsychological test

performance at different age levels in a sample of 182 children with learning problems.⁶ It was found that, for young children (6-8 years old, $n = 90$), there was no significant association between hyperactivity/attentional problems and performance on neuropsychological tasks thought to contain an attentional component. However, for older children (9-12 years old, $n = 92$), there were significant and large negative correlations between a behavioral rating of hyperactivity/attentional problems and performance on attentional measures. These results provide limited support for the maturational lag hypothesis that, although young children with attentional problems may not exhibit deficits relative to normal same-age peers on tasks requiring attention (perhaps because these peers are, themselves, at a low level of attentional development), as they grow older, they may show increasing impairment relative to peers because their attentional abilities are developing at a slower than normal rate. Furthermore, this impairment may not be simply a "delay," which implies an eventual catch-up to peers, but it may be a continuing or increasing deficit. This possibility would need further investigation with older ADD individuals.

In regard to the long-term consequences of ADD, it has been noted that, although hyperactivity and inattention do not appear to worsen with age, the number and severity of their consequences often do increase with age.² ADD chil-

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dren may have an increased risk for developing antisocial behavior,⁷ poor self-esteem,⁸ and depression.⁹

Learning disabilities, frequently defined as a significant deficit in academic achievement in one or more areas despite adequate intellectual abilities,¹⁰ have been found to be quite prevalent among children with ADD.¹¹ Using a regression equation, Cantwell and Satterfield¹¹ found that as many as 76% of the hyperactive children in their study showed significant deficits in at least two academic subjects.

A commonly used method to diagnose learning disabilities is a simple discrepancy criterion. Thus, an individual exhibiting a significant (i.e., one standard deviation or ≥ 15 points) difference between a measure of intellectual ability and academic achievement is diagnosed as having a learning disability. The rationale for this convention is that individuals should normally demonstrate academic skills which are congruent with their intellectual abilities. There is a great deal of controversy regarding the validity of this criterion, but it is still very commonly employed as an "objective" criterion by many school districts in determining eligibility for special education services.¹² One purpose of the current study was to examine the applicability of this simple discrepancy criterion to a sample of ADD children. If it was found that a greater percentage of older ADD children were meeting the discrepancy criterion, one might suspect that younger ADD children may also have a covert learning disability, but one that becomes apparent only over time, as they fall further behind peers in the acquisition of academic skills.

Also, it has been found that, as the ADD child grows older, there is a compounding of attentional, social, and school-related problems which has an increasingly negative impact on academic and behavioral/emotional functioning.¹³ Based on past research suggesting progressive developmental delays in attentional functioning,⁶ it was predicted that, as the ADD child grew older, there would be a significant trend toward decreased academic achievement, including identified learning disabilities and increased behavioral/emotional problems. Specifically, it was hypothesized that there would be a significant negative relationship between age and academic achievement, with older ADD children performing more poorly, with respect to peers, than younger ADD children. Also, it was hypothesized that there would be a significant positive relationship between age and behavioral/emotional problems, with older ADD children obtaining higher scores on a measure of behavioral problems.

METHODS

Subjects

The sample included 79 male children, ranging in age from 6 years, 0 months, to 12 years, 11 months (mean = 8 years, 9 months), selected from a larger sample of 318 children referred to the Austin Neurological Clinic over a 7-year period. The majority of these children were pediatrician-referred for neuropsychological evaluation due to problems in school performance, with teacher reports of difficulties in concentration, staying on-task, and completing assignments. Many of the older subjects were reported

by parents to have had these difficulties since beginning school.

There were 73 Caucasian and six black children. Handedness was assessed using the Harris Tests of Lateral Dominance.¹⁴ There were 71 right-hand preferred, 4 left-hand preferred, and 4 mixed-hand dominant subjects. Four of these children were eliminated from the statistical analyses because of insufficient data.

Each child had been independently diagnosed as having attention deficit disorder (ADD) with or without hyperactivity, using data gathered from parent interviews, direct clinical observations, Conner's Teacher Rating Scale,² the School Situations Questionnaire for Teachers,² and an adapted behavioral rating scale for parents, which included the DSM-III¹⁵ criteria for ADD in questionnaire format. The other 239 children were excluded from the study because of failure to meet the DSM-III criteria or for failure to meet the following criteria: (1) normal corrected or uncorrected vision, (2) normal hearing, (3) full scale IQ (FSIQ) of 80 or above on the Wechsler Intelligence Scale for Children-Revised (WISC-R), or (4) no diagnosis of emotional disturbance, seizures, or past institutional care. Thus, even though a cross-sectional design was employed, homogeneity in subjects across ages was attempted in the selection process.

Measures

All children were administered a battery of psychoeducational tests which included the WISC-R¹⁶ and the Wide Range Achievement Test (WRAT).¹⁷ The following tests from the battery were utilized in the data analysis of the present study: the Full Scale, Verbal, and Performance IQ scores from the WISC-R, and the Reading, Spelling, and Arithmetic subtests from the WRAT. In addition, the parents of each child completed the Child Behavior Checklist (CBC),¹⁸ which is a standardized measure of the child's behavior/personality from the parents' perspective. The scales for boys from this checklist were included in the analyses: the Schizoid, Depression, Somatic Complaints, Uncommunicativeness, Obsessive/Compulsive, Social Withdrawal, Hyperactivity, Aggression, and Delinquency scales.

RESULTS

Descriptive statistics were obtained for the variables included in the study. Means and standard deviations for scores on the Wechsler Intelligence Scale for Children-Revised (WISC-R), the Wide Range Achievement Test (WRAT), and the Child Behavior Checklist (CBC) are listed in Table 1.

The relationship between age and behavioral variables (CBC scales), and age and academic functioning in this sample of attention deficit disorder (ADD) children was examined using multiple correlation coefficients (Table 2). A multiple regression analysis was performed, with age as the dependent variable and the difference scores between Full Scale IQ and the WRAT subtests (FSIQ-WRAT) as the independent variables (an alpha level of ≤ 0.05 was selected as the criterion for significance). The multiple correlation (R) for this set of variables was 0.42 ($p < 0.003$), indicating that 18% ($R^2 = 0.18$) of the variance in age could be

TABLE 1. Mean Performance of Attention Deficit Disorder Children^a

Variable	All (<i>n</i> = 74)	Age Groups	
		6;0–8;5 (<i>n</i> = 38)	8;6–12;11 (<i>n</i> = 36)
Age (mo)	106.8 ± 21.7	88.8 ± 6.9	124.8 ± 15.0
<i>Intellectual^b and Achievement^c Scales</i>			
Full Scale IQ	105 ± 14.4	106 ± 15	105 ± 14
Verbal IQ	104 ± 15.2	104 ± 16	103 ± 15
Performance IQ	106 ± 13.5	106 ± 14	106 ± 12
WRAT-Reading	104 ± 15.5	104 ± 13	104 ± 18
WRAT-Spelling	100 ± 15.7	102 ± 14	99 ± 18
WRAT-Arithmetic	95 ± 13.1	99 ± 13	91 ± 12
<i>Child Behavior Checklist^d</i>			
Aggression	67 ± 10.4 ^e	66 ± 9.7	68 ± 11.0
Delinquency	65 ± 8.7	63 ± 7.2	67 ± 9.5
Depression	64 ± 7.8	62 ± 6.3	66 ± 9.1
Hyperactivity	71 ± 7.6 ^f	69 ± 7.5	73 ± 8.1
Obsessive-Compulsive	64 ± 8.7	63 ± 7.8	66 ± 9.9
Schizoid	63 ± 8.6	62 ± 8.3	64 ± 8.6
Social Withdrawal	64 ± 7.9	62 ± 7.2	68 ± 7.3
Somatic Complaints	62 ± 8.2	61 ± 6.7	64 ± 9.3
Uncommunicative	64 ± 9.5	61 ± 7.4	66 ± 10.0

^aResults are expressed as mean ± SD.

^bWechsler Intelligence Scale for Children-Revised.¹⁶

^cWide Range Achievement Test-Reading, Spelling, and Arithmetic¹⁷ (WRAT).

^dChild Behavior Checklist.¹⁸

^e≤5% of the population.

^f≤2% of the population.

accounted for by FSIQ-WRAT Reading, FSIQ-WRAT Spelling, and FSIQ-WRAT Arithmetic analyzed together. We then performed univariate correlations to examine the relationship between age and the individual academic variables. A significant correlation was found between age and FSIQ-WRAT Arithmetic ($r = 0.35$, $p = 0.002$).

The significant correlation between age and the FSIQ-WRAT Arithmetic discrepancy was then analyzed further. A median split was used to define an older group and a younger group, which were compared for the number of ADD children in each who met or exceeded the 15-point IQ achievement discrepancy commonly used to define learning disability. The younger group consisted of 38 children, ages 6 years, 0 months, to 8 years, 5 months, and the older group consisted of 36 children ages 8 years, 6 months, to 12 years, 11 months. It was found that 29% of the younger children had a 15-point or greater discrepancy, whereas 53% of the older children met this criterion ($\chi^2 = 4.36$, $df = 1$, $p = 0.037$) (Table 3).

In terms of personality/behavioral information, the set of CBC variables was found to have a significant multiple correlation with age ($R = 0.55$, $p = 0.01$) which accounted for 30% of the variance in age ($R^2 = 0.30$) (Table 2). Furthermore, significant univariate correlations were found between age and CBC Social Withdrawal ($r = 0.38$), and Uncommunicative ($r = 0.25$) scales. There were trends toward significance ($p < 0.10$) on the CBC scales of Delinquency ($r = 0.21$), Depression ($r = 0.23$), and Hyperactivity ($r = 0.22$). It also should be noted that this sample of children had mean scores at or above the 95th percentile on the CBC scales of Hyperactivity and Aggression, and

all CBC scores were consistent with a worsening of behavioral problems with increasing age (Table 1).

DISCUSSION

It is recognized that one of the major limitations of the current study is its cross-sectional design. For this reason, the findings must be considered preliminary. Yet, it is felt that these results can be a useful heuristic to guide future longitudinal research on the evolving developmental effects of attention deficit disorder (ADD) on academic and behavioral/emotional functioning.

The results of this study indicated that, in a sample of children diagnosed with ADD, there was a significant negative relationship between age and a measure of arithmetic skill, but there was no significant relationship between age and measures of reading recognition and spelling. With the above-mentioned design limitations in mind, these results suggest that as ADD children get older, they may fall farther behind peers in math achievement. However, the acquisition of reading recognition and spelling skills may be less negatively affected by deficits in the attentional system. For example, a minor attentional error in arithmetic calculation, such as failing to carry, in addition, or not cancelling, in subtraction, will lead to an incorrect answer. In addition, acquisition of arithmetic skills is cumulative, and a failure to acquire early skills may have progressively deleterious effects. In contrast, the acquisition of individual words in spelling and reading may be less of a conceptually cumulative process. Once the child has acquired the concept of sound/symbol matching, adding new words to the reading

TABLE 2. Correlations for Intellectual and Achievement-Difference Scores and the CBC^a with Age

Variable	Pearson	<i>p</i>
<i>Difference scores^b (n = 74)</i>		
FSIQ-WRATR	0.01	0.95
FSIQ-WRATS	0.12	0.30
FSIQ-WRATA	0.35	0.002
Overall test of significance: $R = 0.42^c$, $p = 0.003$		
<i>Child Behavior Checklist (n = 67)</i>		
Aggression	0.09	0.45
Delinquency	0.21	0.09
Depression	0.23	0.06
Hyperactivity	0.22	0.07
Obsessive-Compulsive	0.05	0.68
Schizoid	0.13	0.30
Social Withdrawal	0.38	0.002
Somatic Complaints	0.11	0.36
Uncommunicative	0.25	0.04
Overall test of significance: $R = 0.55^c$, $p = 0.01$		

Two-tailed test of significance.

^aChild Behaviors Checklist.¹⁸^bFull Scale IQ¹⁶ minus the Wide Range Achievement Test—Reading, Spelling, and Arithmetic.¹⁷^cMultiple Correlation.

and spelling vocabulary may be a matter of rote memorization, whereas mathematic development may involve the acquisition of more complex concepts and procedures. Thus, the older the ADD child becomes, the more s/he may begin to lag behind in arithmetic skills. Also, in a more general sense, as the ADD child experiences increasing school failure and frustration, s/he may lose motivation and invest less and less effort in learning. Furthermore, the emotional concomitants of ADD, also shown to increase with age in this study, may exacerbate academic difficulties.

In terms of intellectual and academic skills examined in the current study, there were no differences, other than arithmetic scores, between younger and older children. Given that only 18% of the total variance in age was accounted for by the academic variables examined, additional research is needed to evaluate other academic and cognitive variables. For example, it could be important to examine other areas of verbal achievement in ADD children, such as reading comprehension and written expression. These tasks involve more complex verbal academic skills than single word-reading recognition and spelling, and may be more negatively affected by attentional deficits and cognitive impulsivity. Informal clinical observations indicate that written expression is often an area of deficit for ADD children. For example, handwriting is often poor, punctuation and capitalization may be deficient, organization is often lacking, and the child may misspell words during paragraph writing that were correctly spelled on the Wide Range Achievement Test.¹⁷ Of course, empirical evidence is needed to confirm these clinical observations.

Correlational results between age and academic variables were examined through the application of a simple discrepancy criterion. In the past, a 15-point discrepancy between IQ and academic achievement has been used widely as the

TABLE 3. Percentage of Cases with ≥ 15 -point Discrepancy on FSIQ^a and WRAT-A^b Performance across Age

	Age		Total
	Young 6;0–8;5	Old 8;6–12;11	
<15 point	71.1%	47.2%	59.5%
≥ 15 point	28.9%	52.8%	40.5%
Total	51.4%	48.6%	100%

 $\chi^2 = 4.36$ (1), $p = 0.037$ ^aFull Scale IQ¹⁶^bWide Range Achievement Test—Arithmetic.¹⁷

educational criterion for diagnosing learning disability (LD), and thus qualifying the child for special education services.¹² In the current investigation, it was found that 41% of these ADD children met the discrepancy criterion for LD. However, the results also indicated that younger ADD children were less likely to be diagnosed as having LD than older ADD children, when a discrepancy formula is used. One possible explanation for these results may be that there are a number of ADD children whose learning difficulties remain undiagnosed until they have been in school long enough for a substantial gap to develop between IQ and academic skills. Using only a simple 15-point discrepancy criterion to identify LD in the younger child may delay diagnosis and intervention, possibly leading to adverse emotional and educational repercussions. Perhaps the ADD child could be identified as being at risk for the development of LD, and some type of early intervention could then be implemented as a preventative measure, before a substantial discrepancy develops. Also, a more sophisticated means of diagnosing LDs may be more appropriate, such as applying a regression equation which includes chronological age and academic and cognitive variables. Future research comparing the efficacy of these two methods with a sample of ADD children would be quite beneficial.

With regard to emotional/behavioral functioning, this sample of ADD children had mean scores above the 95th percentile on Child Behavior Checklist (CBC) scales related to aggression and hyperactivity. In addition, older ADD children were found to be significantly more uncommunicative and socially withdrawn than were younger ADD children. These findings indicate that, regardless of age, ADD children are perceived as being aggressive and abrasive in social relations. One can further hypothesize that, as the ADD child becomes older, this acting-out style may lead to social rejection and alienation from others which, in turn, may lead to increasing problems with social withdrawal and uncommunicativeness. A longitudinal study would allow a test of this inference based on the present cross-sectional study. Also, although there was a significant multiple correlation between age and CBC variables ($R = 0.55$), this accounted for only 30% of the variance in age. This finding indicates that it will be very important to examine other emotional/behavioral variables in the future, such as self-esteem and social skills.

In conclusion, the results from the present investigation support the proposal that, as the ADD child grows older,

there are increasing negative repercussions associated with ADD in some areas of academic and behavioral functioning. Specifically, in this sample of ADD children, there was a significant relationship between age and poorer math performance, social withdrawal, and uncommunicativeness. It

is recognized that the generalizability of findings from the current study is limited by the exclusion of females from the sample. Research is ongoing at the present time to determine whether these findings apply also to female ADD children.

REFERENCES

1. American Psychiatric Association: Diagnostic and Statistical Manual of Mental Disorders (Third Edition-Revised). Washington, DC, 1987
2. Barkley RA: Hyperactive Children: A Handbook for Diagnosis and Treatment. New York, Guilford Press, 1981
3. Chelune GJ, Ferguson W, Koon R, et al: Frontal lobe disinhibition in attention deficit disorder. *Child Psychiatry Hum Dev* 16:221-234, 1986
4. Yakovlev PI, Lecours AR: The myelogenetic cycles of regional maturation of the brain, in Minkowski A (ed): *Regional Development of the Brain in Early Life*. Oxford, Blackwell Press, 1967
5. Picton TW, Campbell KB, Baribeau-Braun J, et al: The neurophysiology of human attention: A tutorial review, in Requin J (ed): *Attention and Performance*. Hillsdale, NJ, Erlbaum, 1978
6. Massman PJ, Nussbaum NL, Bigler ED: The mediating effect of age on the relationship between Child Behavior Checklist hyperactivity scores and neuropsychological test performance. *J Abnorm Child Psychol* 16:89-95, 1988
7. Satterfield JH, Hoppe CM, Schell AM: A prospective study of delinquency in 110 boys with attention deficit disorder and 88 normal adolescent boys. *Am J Psychiatry* 139:795-798, 1982
8. Campbell SB, Endman MW, Bernfeld, G: A three-year follow-up of hyperactive preschoolers into elementary school. *J Child Psychol Psychiatry* 18:239-249, 1977
9. Brumback, RA, Weinberg WA: Relationship of hyperactivity and depression in children. *Percept Mot Skills* 45:247-251, 1977
10. Gaddes WH: *Learning Disabilities and Brain Function; A Neuropsychological Approach*. New York, Springer-Verlag, 1983
11. Cantwell DP, Satterfield JH: The prevalence of academic underachievement in hyperactive children. *J Pediatr Psychol* 3:168-171, 1978
12. Chalfant JC: *Identifying Learning Disabled Students: Guidelines for Decision Making*. Andover, MA: Network of Innovative Schools. (ERIC Document Reproduction Service No. ED 258 390), 1984
13. Weiss G, Hechtman LT: *Hyperactive Children Grown Up*. New York, Guilford Press, 1986
14. Harris AJ: *Harris Tests of Lateral Dominance, manual of directions for administration and interpretation*. New York, Psychological Corp, 1947
15. American Psychiatric Association: *Diagnostic and Statistical Manual of Mental Disorders* (3rd ed.). Washington, DC, 1980
16. Wechsler D: *Wechsler Intelligence Scale for Children-Revised*. New York, Psychological Corporation, 1974
17. Jastak J, Jastak S: *The Wide Range Achievement Test manual*. Wilmington, DE, Guidance Associates, 1965
18. Achenbach T, Edelbrock C: *The Child Behavior Checklist-Revised*. United States, Queen City Printers, 1983

ANNOUNCEMENT

MIAMI CHILDREN'S HOSPITAL ELEVENTH ANNUAL CHILD NEUROLOGY POSTGRADUATE COURSE June 11-14, 1990, to be held at the Sonesta Beach Hotel, Key Biscayne, Florida. Credit Category I credit hours. *Topics:* neonatal neurology, developmental disorders, neuromuscular diseases, and seizures. For more information contact: Oscar Papazian, M.D., Miami Children's Hospital, Department of Neurology, 3200 S.W. 60th Ct., Miami, FL 33155. (305) 662-8330.